

Newtrinos.jl: A Julia Package for Global Analysis of Neutrino Data

Philipp Eller¹ and David Schultheiß¹

¹ Technical University of Munich, Germany

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

- [Review](#)
- [Repository](#)
- [Archive](#)

Editor: [Matthew Feickert](#)

Reviewers:

- [@rafaelab](#)
- [@szabo137](#)

Submitted: 28 October 2025

Published: unpublished

License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](#)).

Summary

Newtrinos.jl is an open-source Julia package for performing global analyses of neutrino data. It provides a modular, three-layer architecture that separates physics models, experiment descriptions, and statistical inference into independent modules. This allows researchers to freely combine experiments and test them against a variety of theoretical models. New experiments and physics models can be added without modifying core code.

Full statistical forward models including all relevant systematic uncertainties are implemented for each experimental dataset, defining both the likelihood and the data-generating process and enabling modern statistical inference workflows.

The framework is composable, making it straightforward to construct joint likelihoods over multiple datasets. Physics and nuisance parameters can be merged, correlated, or decorrelated across experiments to ensure consistency in the joint fit.

The package supports frequentist and Bayesian inference and is parallelisable across CPU threads or distributed workers. As the package is written entirely in Julia, all models are automatically differentiable, enabling exact gradient computation.

Statement of Need

As neutrino physics transitions into an era of high-precision measurements, global fits have become essential for extracting neutrino properties and probing physics beyond the Standard Model ([Capozzi et al., 2025](#); [Esteban et al., 2024](#); [Salas et al., 2021](#)). By combining disparate datasets, they break parameter degeneracies that limit single-experiment analyses. Consequently, they are poised to address urgent open questions such as the neutrino mass ordering and CP violation in the lepton sector. The first joint fit of T2K and NOvA data demonstrated that combining distinct experimental signatures is crucial to resolving such degeneracies and tensions ([Abubakar & others, 2025](#)).

Performing a global neutrino fit requires assembling diverse, experiment-specific components, such as detector models, neutrino flux models, matter density profiles, and interaction cross sections, into a coherent inference pipeline, while rigorously accounting for large numbers of correlated systematic uncertainties. Varying availability of public data and documentation further limits dataset compatibility, making global fits a substantial undertaking that can render certain analyses infeasible ([Capozzi et al., 2025](#)). Existing frameworks that have enabled successful studies are typically proprietary and closed-source, making results difficult to verify and the software impossible to reuse or adapt.

A further challenge is the computational burden of high-dimensional parameter spaces with correlated systematics. As the number of experiments grows, traditional statistical methods become increasingly intractable. Automatic differentiation and gradient-based inference offer

promising paths forward. However, existing frameworks (see [Related Work](#)) were developed before these techniques became widely adopted and therefore rely primarily on derivative-free optimisation or finite-difference gradient estimates such as those used by MINUIT ([James & Roos, 1975](#)).

Newtrinos.jl addresses these limitations with a fully open-source, extensible framework built within the high-performance Julia ecosystem.

Key Features

- **Out-of-the-box usability:** The package includes experimental data, configuration files, and plotting tools, making it straightforward to reproduce results, verify correctness, and conduct custom analyses on existing models and experiments.
- **Modular architecture:** Experiments, physics models, and analysis methods are implemented as independent components with clearly separated concerns. The physics layer provides theory predictions and physical parameters; the experiment layer configures individual experiments with their parameters, priors, and forward models; the analysis layer provides inference methods for sampling, profiling, or scanning. Each layer treats the others as black boxes, enabling flexible composition. Layers communicate through two well-defined interfaces: `NamedTuples` of parameters and priors adhering to the `Distributions.jl` standard, and callable functions stored in structs. Theory extensions can be added via Julia's multiple dispatch without modifying existing code, and parameters flow through the pipeline automatically.
- **Full statistical models:** The package constructs full statistical forward models for each experiment, encoding the complete mapping from physical parameters to expected observations. Systematic uncertainties enter as nuisance parameters rather than being absorbed into simplified χ^2 approximations, preserving the full statistical information. This enables profiling or marginalisation of joint likelihoods, Bayesian posterior estimation, and the generation of pseudo-data and prior and posterior predictive checks.
- **Consistent parameter handling:** All physics and nuisance parameters across configured experiments are collected into a single, flat `NamedTuple` via `get_params` and `get_priors` with `safe_merge`. Wrapper functions support arbitrary parameter renaming and (de)correlation between experiments. Prior distributions, including correlated priors via covariance matrices, are managed in the same unified structure and compose naturally into a joint prior. Individual parameters can be overridden or conditioned at analysis time without modifying the underlying modules, keeping configurations transparent and reproducible.
- **Automatic differentiation:** The complete forward model chain — from oscillation probability calculation to likelihood evaluation — is compatible with `ForwardDiff.jl` out of the box, with `Mooncake.jl` also successfully tested. Exact, efficient gradients enable advanced optimisation and inference methods such as those provided by `BAT.jl` ([Schulz et al., 2021](#)).
- **Scalability:** The package is built for large-scale inference tasks with tens of experiments and hundreds of parameters. Julia's just-in-time compilation delivers high performance without sacrificing expressiveness. Profile likelihood scans parallelise transparently across CPU threads or distributed workers via `Distributed.jl`, and forward-mode automatic differentiation keeps gradient-based inference tractable as the parameter space grows.

84 Workflow Overview

85 Newtrinos.jl is designed as a modular pipeline where components communicate strictly through
86 functional interfaces: physics modules expose callables used inside experiment forward models,
87 and experiments expose forward model callables consumed by the analysis layer, without either
88 inspecting the other's internals. A typical analysis proceeds as follows:

- 89 1. **Configure physics:** Select and instantiate theoretical models — for example, a three-
90 flavour oscillation model with non-standard matter interactions, an Earth density model,
91 an atmospheric flux model, and a cross-section model. Any model can be swapped
92 independently, e.g. replacing a three-flavour model with a BSM model including sterile
93 neutrinos.
- 94 2. **Configure experiments:** Select one or more experiment modules, each encapsulating its
95 detector response, systematic uncertainties, and observed data. Instantiate them via the
96 configure method with the chosen physics model. Experiments are independent and
97 can be freely combined, typically sharing a common physics model.
- 98 3. **Build the joint likelihood:** Pass the collection of configured experiments to
99 generate_likelihood, which composes their individual forward models and likelihood
100 functions into a single joint likelihood.
- 101 4. **Collect parameters and priors:** Use get_params and get_priors to collect and merge
102 all physics and nuisance parameters across experiments into a unified NamedTuple.
103 Parameters and priors can be modified as needed, e.g. for likelihood conditioning via
104 Accessors.jl.
- 105 5. **Run inference:** Choose an analysis method from the provided tools. For example, run a
106 profile likelihood scan by calling profile on the joint likelihood over a chosen parameter
107 grid to obtain a NewtrinosResult containing the grid coordinates, likelihood values,
108 optimised nuisance parameters at each point, and run metadata.
- 109 6. **Visualise and export results:** Plot confidence contours or best-fit data/MC comparisons
110 using the built-in plotting utilities based on Makie.jl, and save results to disk.

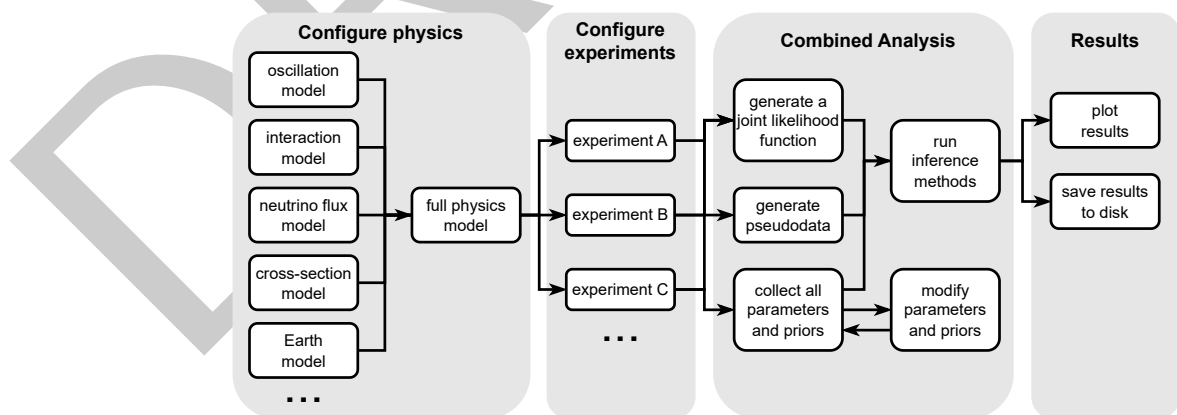


Figure 1: Typical workflow for global analyses of neutrino data with Newtrinos.jl. The layers communicate through two well-defined interfaces: NamedTuples of parameters and priors, and callable functions stored in structs.

111 Related Work

112 Several existing software projects address related but distinct use cases:

- 113 ■ **GLOBES** (Huber et al., 2005, 2007): Simulates long-baseline experiments but does not
- 114 support full global fits.
- 115 ■ **GAMBIT** (Athron & others, 2017): A general-purpose global fitting framework with
- 116 some support for neutrino data, but not tailored for neutrino physics.
- 117 ■ **PEANUTS** (Gonzalo & Lucente, 2024): Focused on solar neutrino modelling.
- 118 ■ **PhyLiNO** (Hellwig et al., 2025): A high-performance framework for reactor neutrino
- 119 data.
- 120 ■ **PISA** (Aartsen & others, 2020): Designed for atmospheric neutrino analyses.
- 121 ■ **MANGO** (Granger, 2026): Does not contain any experimental likelihoods or forward
- 122 models, but provides oscillation probability calculations with automatic differentiation
- 123 support.

124 Newtrinos.jl complements these efforts by focusing on the neutrino sector, offering a simple,
 125 extensible, and efficient design. It is currently the only framework in this domain supporting
 126 automatic differentiation. The software has been used in in multiple research publications
 127 (Eller et al., 2025; Eller, 2026; Ettengruber et al., 2024; Kozynets et al., 2025; Lonardi et al.,
 128 2026).

129 Acknowledgements

130 This work was supported by Germany's Federal Ministry of Research, Technology and
 131 Space (BMFTR) within the ErUM-Data programme under grant FKZ 05D25PC1 (DEMOS
 132 consortium), and partially by the Deutsche Forschungsgemeinschaft (DFG, German Research
 133 Foundation) under Germany's Excellence Strategy – EXC-2094/2 – 390783311, the SFB 1258
 134 – 283604770, and NFDI 39/1.

135 Aartsen, M. G., & others. (2020). Computational techniques for the analysis of small signals
 136 in high-statistics neutrino oscillation experiments. *Nucl. Instrum. Meth. A*, 977, 164332.
 137 <https://doi.org/10.1016/j.nima.2020.164332>

138 Abubakar, S., & others. (2025). Joint neutrino oscillation analysis from the T2K and NOvA
 139 experiments. *Nature*, 646, 818–824. <https://doi.org/10.1038/s41586-025-09599-3>

140 Athron, P., & others. (2017). GAMBIT: The Global and Modular Beyond-the-Standard-Model
 141 Inference Tool. *Eur. Phys. J. C*, 77(11), 784. <https://doi.org/10.1140/epjc/s10052-017-5321-8>

143 Capozzi, F., Giarè, W., Lisi, E., Marrone, A., Melchiorri, A., & Palazzo, A. (2025). Neutrino
 144 masses and mixing: Entering the era of subpercent precision. *Phys. Rev. D*, 111(9),
 145 093006. <https://doi.org/10.1103/PhysRevD.111.093006>

146 Eller, P. (2026). Atmospheric neutrino oscillations: The full picture. *Phys. Rev. D*, 114(5),
 147 053005. <https://doi.org/10.1103/v4hl-pbpn>

148 Eller, P., Ettengruber, M., & Zander, A. (2025). Neutrino data analysis of extra-dimensional
 149 theories with massive bulk fields. *Phys. Rev. D*, 112(5), 055009. <https://doi.org/10.1103/1llm-96vy>

151 Esteban, I., Gonzalez-Garcia, M. C., Maltoni, M., Martinez-Soler, I., Pinheiro, J. P., & Schwetz,
 152 T. (2024). NuFit-6.0: updated global analysis of three-flavor neutrino oscillations. *JHEP*,
 153 12, 216. [https://doi.org/10.1007/JHEP12\(2024\)216](https://doi.org/10.1007/JHEP12(2024)216)

154 Ettengruber, M., Zander, A., & Eller, P. (2024). Testing the number of neutrino species with
 155 a global fit of neutrino data. *Phys. Rev. D*, 109(9), 095016. <https://doi.org/10.1103/PhysRevD.109.095016>

157 Gonzalo, T. E., & Lucente, M. (2024). PEANUTS: a software for the automatic computation
 158 of solar neutrino flux and its propagation within Earth. *Eur. Phys. J. C*, 84(2), 119.
 159 <https://doi.org/10.1140/epjc/s10052-024-12423-3>

- 160 Granger, P. (2026). *MANGO: An Autodiff Neutrino Oscillation Engine for Differentiable*
161 *Analysis Pipelines*. <https://arxiv.org/abs/2608.13429>
- 162 Hellwig, D., Schoppmann, S., Soldin, P., Stahl, A., & Wiebusch, C. (2025). PhyLiNO: a
163 forward-folding likelihood-fit framework for neutrino oscillation physics. *Comput. Softw.*
164 *Big Sci.*, 9(1), 11. <https://doi.org/10.1007/s41781-025-00142-7>
- 165 Huber, P., Kopp, J., Lindner, M., Rolinec, M., & Winter, W. (2007). New features in the
166 simulation of neutrino oscillation experiments with GLoBES 3.0: General Long Baseline
167 Experiment Simulator. *Comput. Phys. Commun.*, 177, 432–438. <https://doi.org/10.1016/j.cpc.2007.05.004>
- 168
- 169 Huber, P., Lindner, M., & Winter, W. (2005). Simulation of long-baseline neutrino oscillation
170 experiments with GLoBES (General Long Baseline Experiment Simulator). *Comput. Phys.*
171 *Commun.*, 167, 195. <https://doi.org/10.1016/j.cpc.2005.01.003>
- 172 James, F., & Roos, M. (1975). Minuit: A System for Function Minimization and Analysis
173 of the Parameter Errors and Correlations. *Comput. Phys. Commun.*, 10, 343–367.
174 [https://doi.org/10.1016/0010-4655\(75\)90039-9](https://doi.org/10.1016/0010-4655(75)90039-9)
- 175 Kozynets, T., Eller, P., Zander, A., Ettengruber, M., & Koskinen, D. J. (2025). Constraints
176 on non-unitary neutrino mixing in light of atmospheric and reactor neutrino data. *JHEP*,
177 05, 130. [https://doi.org/10.1007/JHEP05\(2025\)130](https://doi.org/10.1007/JHEP05(2025)130)
- 178 Lonardi, S., Ettengruber, M., & Eller, P. (2026). *Searching for the N -naturalness tower of*
179 *neutrinos*. <https://arxiv.org/abs/2607.14243>
- 180 Salas, P. F. de, Forero, D. V., Gariazzo, S., Martínez-Miravé, P., Mena, O., Ternes, C. A.,
181 Tórtola, M., & Valle, J. W. F. (2021). 2020 global reassessment of the neutrino oscillation
182 picture. *JHEP*, 02, 071. [https://doi.org/10.1007/JHEP02\(2021\)071](https://doi.org/10.1007/JHEP02(2021)071)
- 183 Schulz, O., Beaujean, F., Caldwell, A., Grunwald, C., Hafych, V., Kröninger, K., Cagnina, S.
184 L., Röhrig, L., & Shtembari, L. (2021). BAT.jl: A julia-based tool for bayesian inference.
185 *SN Computer Science*, 2(3), 210. <https://doi.org/10.1007/s42979-021-00626-4>